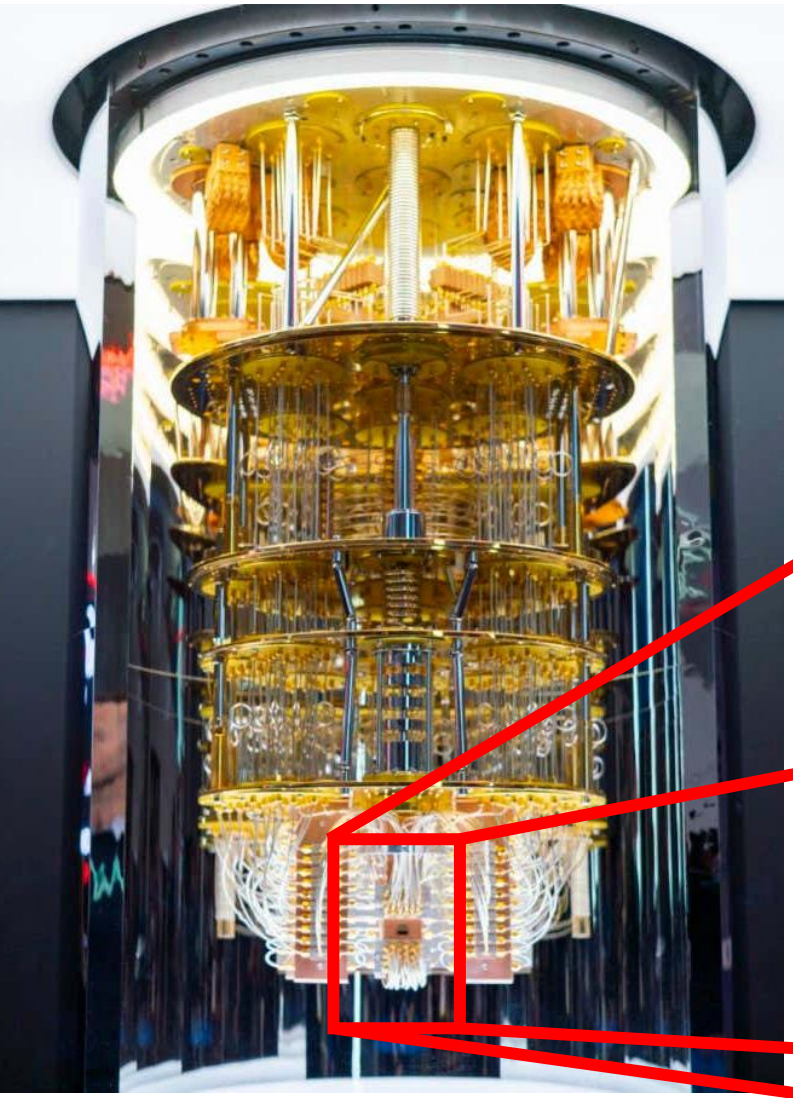
How Do We Read a Superconducting Qubit?

Peiyan Li

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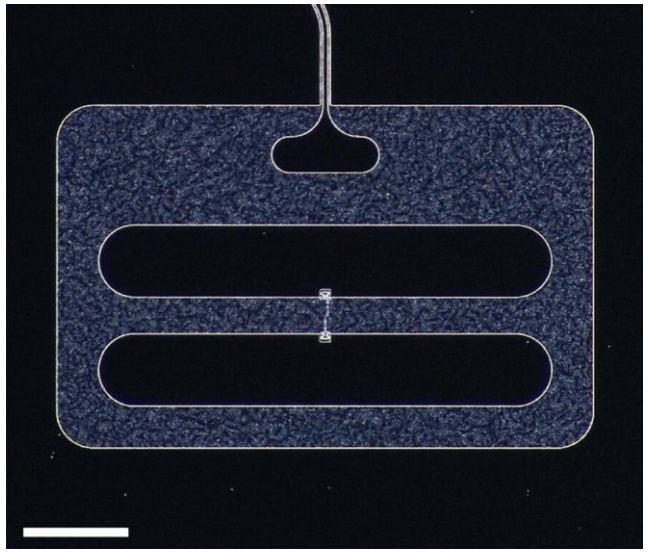
10/23/2026

Qubit State Information Is Encoded in a Microwave Signal



A Superconducting Quantum Computer
Image: newscientist.com

- Superconducting qubit = circuit-based two-level system
- Operates at ~10 mK in a dilution refrigerator
- State information is encoded in a microwave signal
  (Phase Difference)



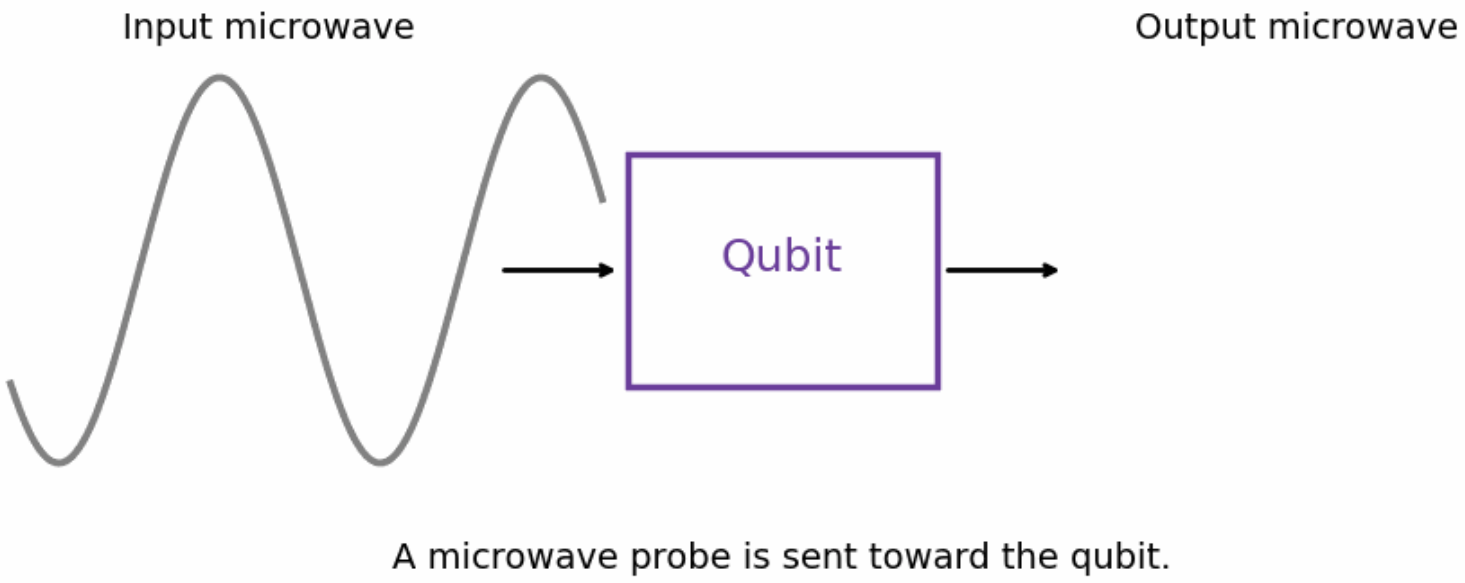
Matthew P. Bland et al.

Qubit State Information Is Encoded in a Microwave Signal



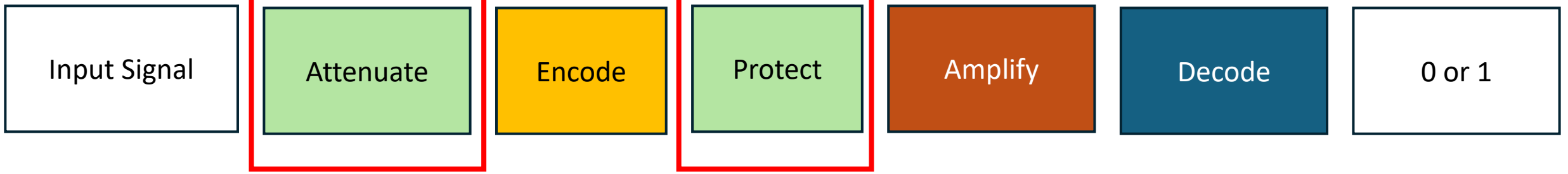
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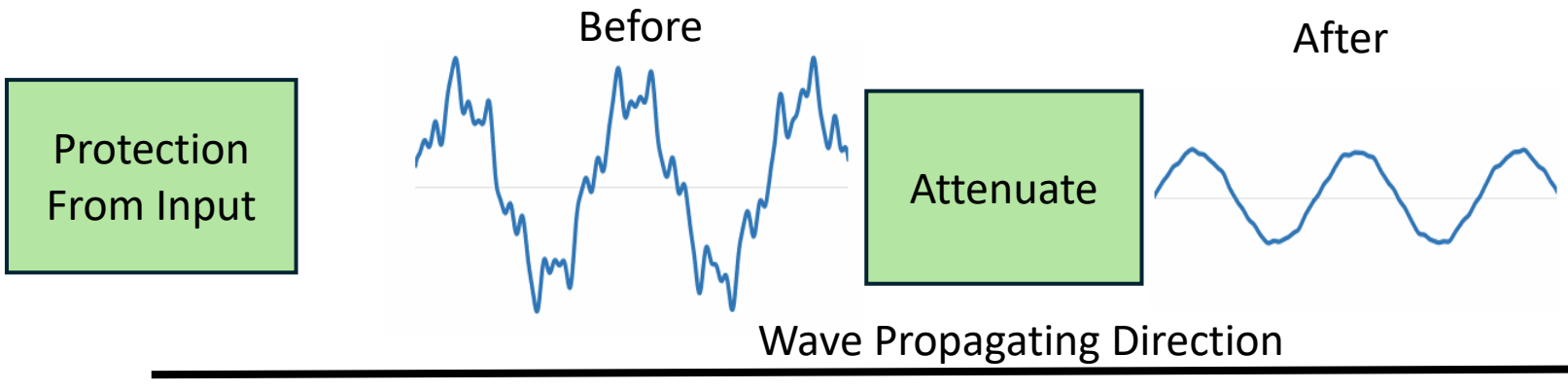
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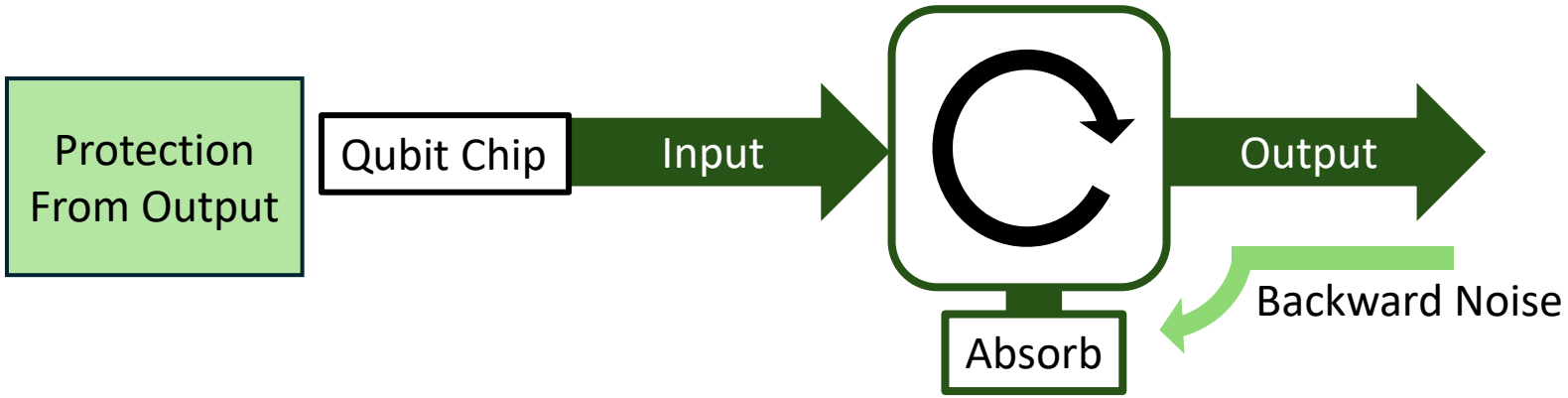
Different Qubit States Produce Different Microwave Phases

How Can We Protect Superconducting Qubit when Reading Out?





Protection:
1. Reduce Probe Power
2. Suppress Thermal Noise



Protection:
1. **One-Way Propagation**
2. **Redirect Backward Noise to Absorber**
3. **Prevent Qubit from Backward Noise**

Schematic of a Protection Process

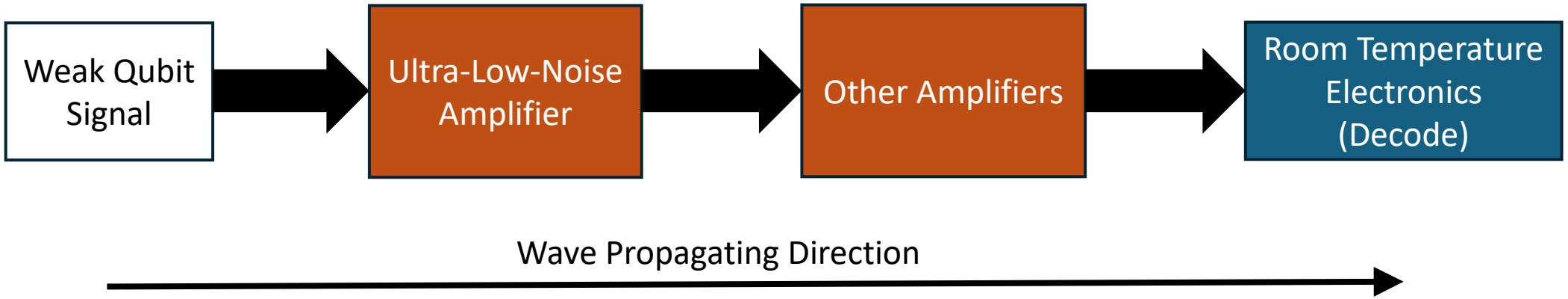
How Can We Amplify Weak Qubit Signals?

Input Signal | Attenuate | Encode | Protect | **Amplify** | Decode | 0 or 1

Amplify

Amplify Signal with Minimum Noise

Key idea: The earliest amplifier must add as little noise as possible.



Fun fact:

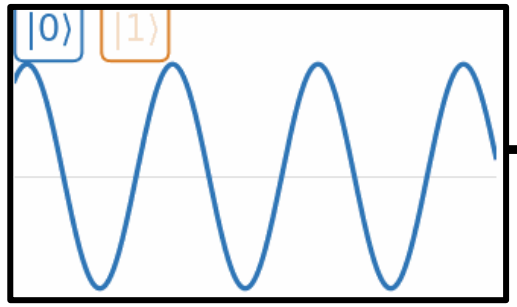
Quantum mechanics sets a fundamental lower limit on added noise.

Ultra-low-noise amplifiers can operate very close to this limit.

Decode state information from microwave signal

Input Signal | Attenuate | Encode | Protect | Amplify | **Decode** | 0 or 1

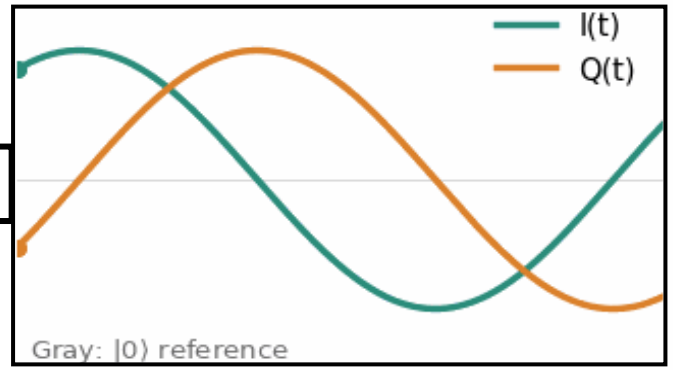
Local Oscillating Wave



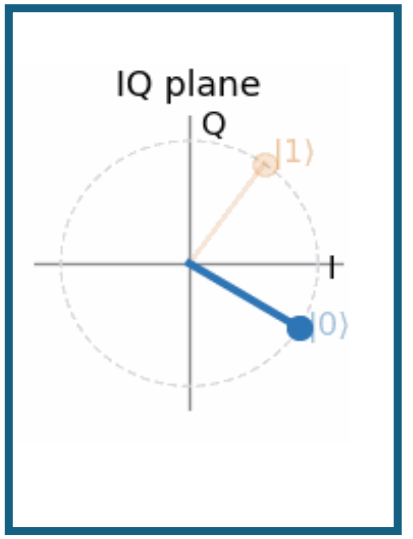
Signal with state information

Generate two components
$I(t)$ $Q(t)$

Low Pass Filter



State-dependent I/Q signals after low-pass filtering.



Phase Different from Qubit States → Distinguishable Qubit State

Schematic of a Decoder (Demodulator)

From Qubit State to Classical Bit



1. Encode

   Qubit state changes the microwave phase

2. Protect

   Protect the qubit from noise

3. Amplify

   Amplify weak signals

4. Decode

   Extract phase information to distinguish 0 and 1

Quantum state information → Microwave phase → Classical measurement

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